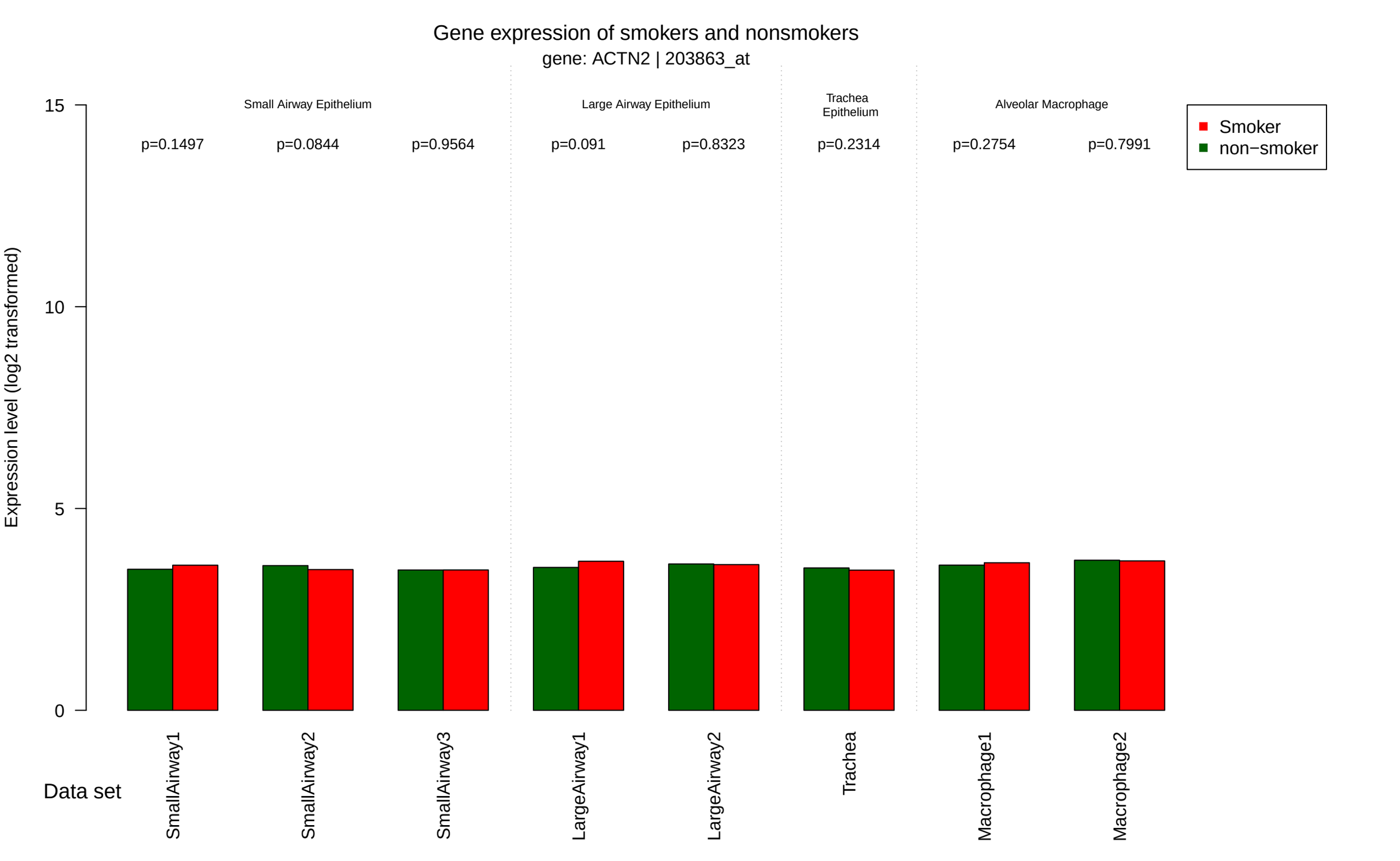
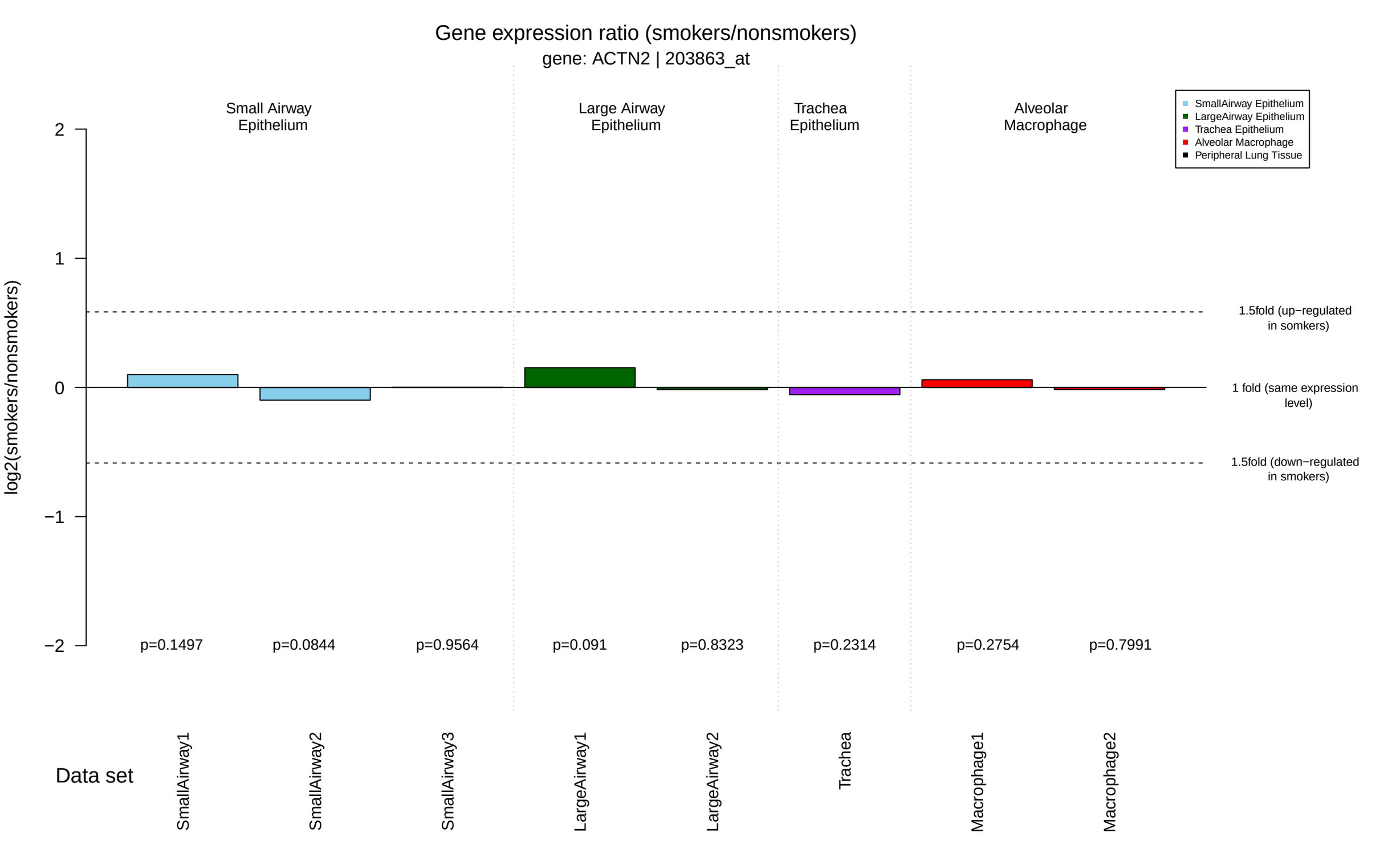
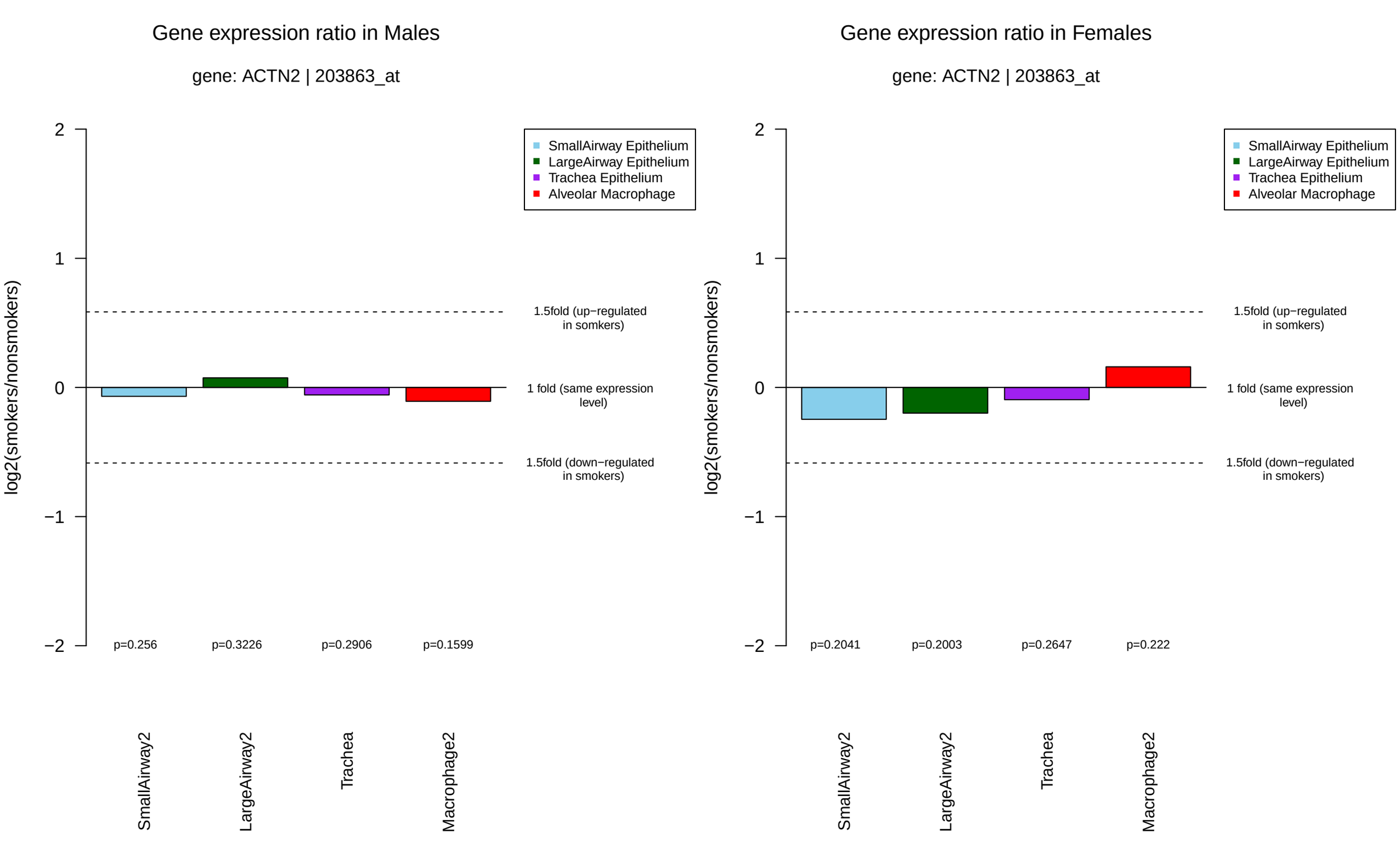
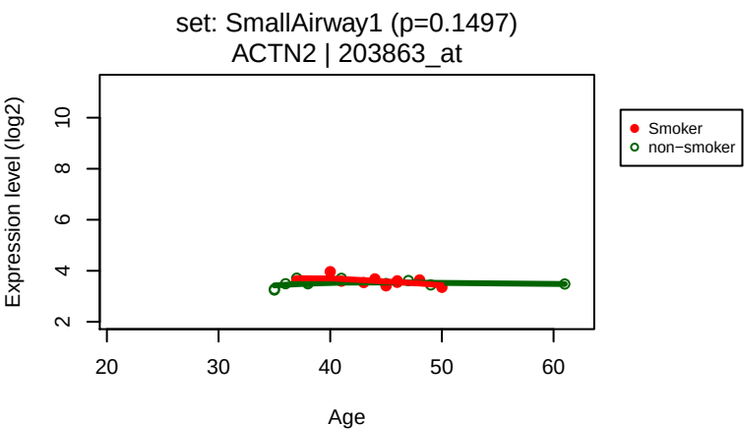
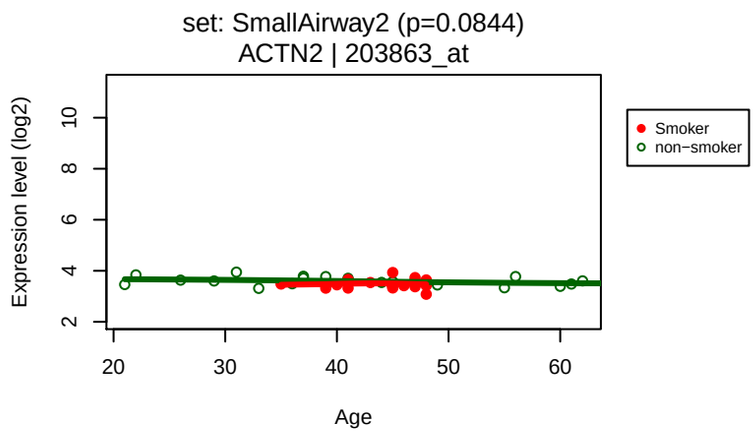
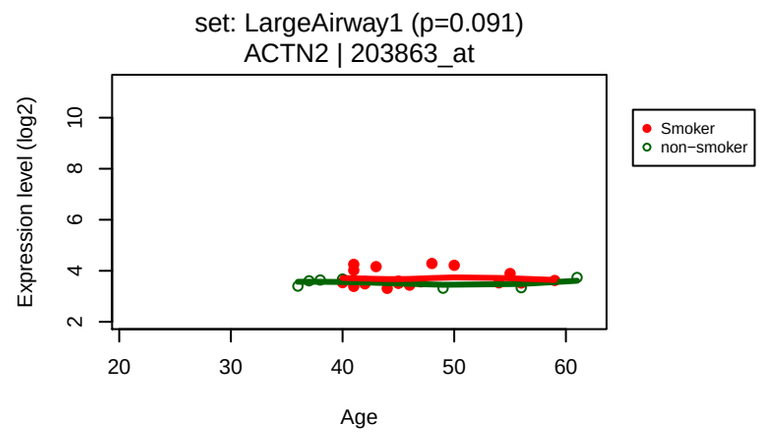

Gene expression of smokers and nonsmokers
gene: ACTN2 | 203863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1497
p=0.0844
p=0.9564
p=0.091
p=0.8323
p=0.2314
p=0.2754
p=0.7991
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

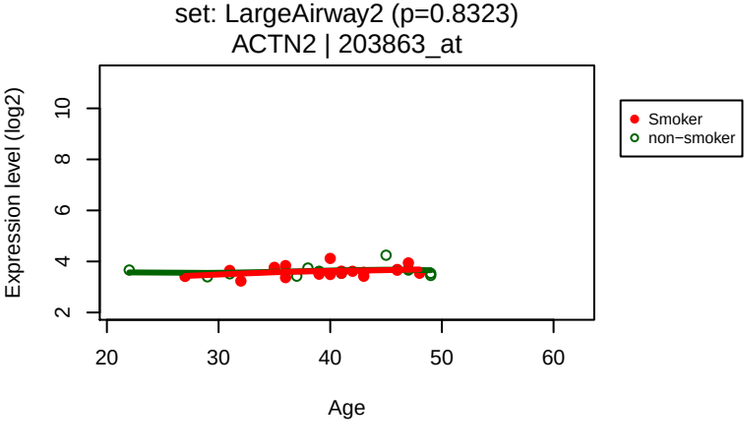

Gene expression ratio (smokers/nonsmokers)
gene: ACTN2 | 203863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1497
p=0.0844
p=0.9564
p=0.091
p=0.8323
p=0.2314
p=0.2754
p=0.7991
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

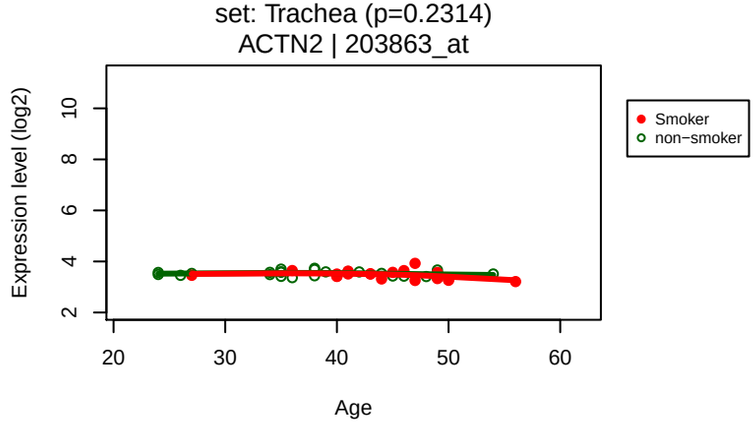

Gene expression ratio in Males
gene: ACTN2 | 203863_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.256
p=0.3226
p=0.2906
p=0.1599
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN2 | 203863_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2041
p=0.2003
p=0.2647
p=0.222
SmallAirway2
LargeAirway2
Trachea
Macrophage2

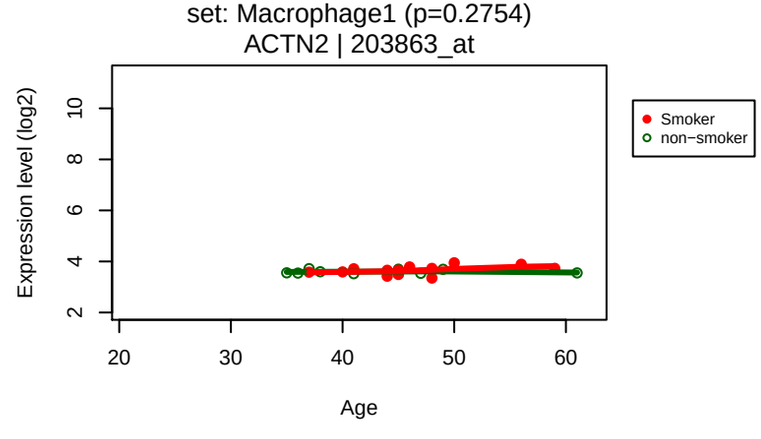

set: SmallAirway1 (p=0.1497)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker

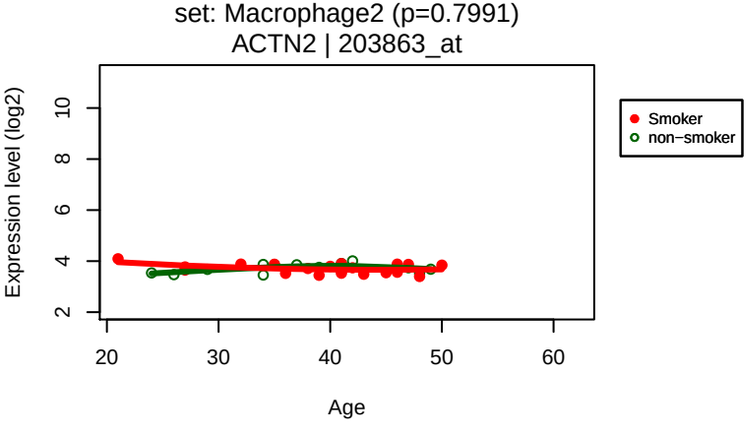

set: SmallAirway2 (p=0.0844)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker

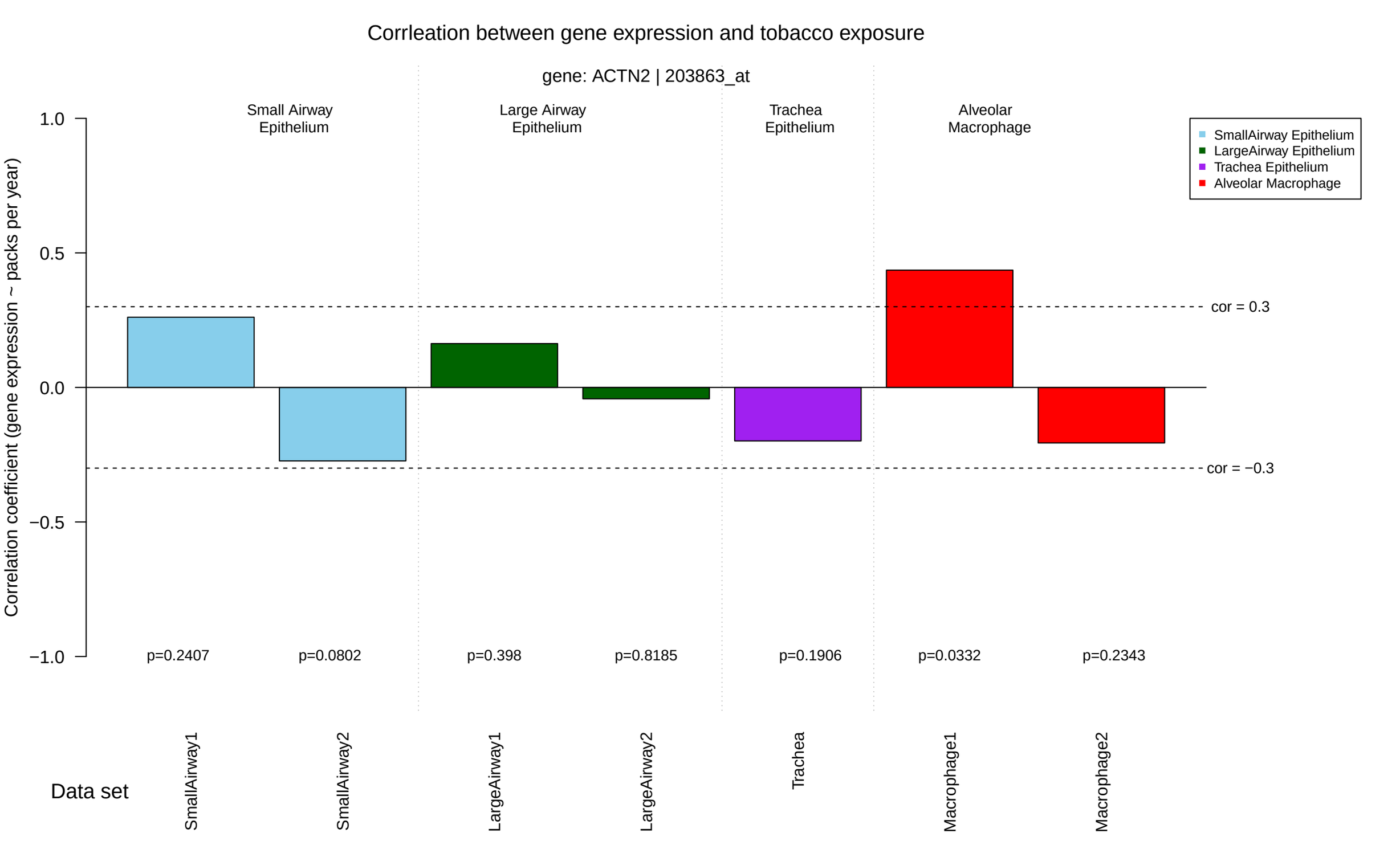

set: LargeAirway1 (p=0.091)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8323)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2314)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2754)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7991)
ACTN2 | 203863_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN2 | 203863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2407
p=0.0802
p=0.398
p=0.8185
p=0.1906
p=0.0332
p=0.2343
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2